Generative AI to Handle Class Imbalance Using Synthetic Data Augmentation

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***Abstract*—In machine learning class imbalance remains a big issue especially in high-risk areas such as fraud or medical diagnosis where the instances from minority classes are rare but important. Despite their high overall accuracy conventional classifiers have a tendency to favour the majority class which leads to poor detection of minority events. With the help of creating the interpolated samples, traditional oversampling techniques, such as Synthetic Minority Over-sampling Technique (SMOTE), can help to address this issue, although it may not be able to capture intricate feature distribution in structured datasets. In this work, we investigate the effectiveness of a synthetic minority class augmentation generative AI technique under extreme imbalance, i.e. Conditional Tabular Generative Adversarial Networks (CTGAN). We use class-aware evaluation metrics such as Recall, F1-score and Precision-Recall Area Under the Curve (PR-AUC) to compare the evaluation of baseline models, SMote based augmentation and CTGAN generated synthetic data using Credit Card Fraud Detection dataset. In comparison to the baseline model, from experimental results, it can be seen that CTGAN achieves better minority class recall which may indicate higher sensitivity towards uncommon things that are occurring or rare. SMOTE, on the other hand, has a higher PR-AUC which denotes a more stable balance between precision and recall. These results show the compromises between the performance all the way from interpolation/oversampling performing, compared to generative augmentation in imbalanced tabular classification tasks.**

***Index Terms*—Class Imbalance, Generative Adversarial Networks, CTGAN, Synthetic Data Augmentation, Fraud Detection, Precision–Recall AUC, SMOTE, Imbalanced Learning**

I. Introduction

In machine learning classification tasks, class imbalance is a common problem especially in industries where less frequent occurrences have significant consequences. Applications in fields such as cyber security, medical diagnosis, fraud detection, fault detection etc often contain datasets where minority class forms a very small fraction of the observations. Even though the minority class is rare, it is typically correlated to significant results, and so accurate detection is important. The majority class is usually favored in conventional classifiers in unbalanced data sets and is likely to lead to misleading high accuracy yet is insufficient to detect the minority instances.

Numerous resampling techniques have been put forth to solve this problem. Conventional techniques for oversampling such as the Synthetic Minority Over-sampling Technique (SMOTE) generate artificial data samples through interpolation between the existing minority samples. Although these methods help to rebalance the dataset, it may not be able to represent complex feature distribution or nonlinear relationships in structured data well. Moreover, overlapping regions of class could be introduced by techniques based on interpolations, which could affect the accuracy.

New opportunities for data augmentation have been created by recent developments in generative artificial intelligence. The aim of Generative Adversarial Networks (GANs), in particularing their type that is GAN for tabular data, Conditional Tabular GAN (CTGAN) is to generate realistic fake data samples by learning the underlying distribution of the data. GANs attempt to model the whole joint feature distribution, unlike the techniques of interpolation GANs, which may lead to more representative and diverse minority samples.

In this work, we are applying the highly unbalanced Credit Card Fraud Detection dataset to carry out a systematic comparison of baseline classification models, SMote-based oversampling and CTGAN-generated synthetic augmentation. We investigate the tradeoffs between the sensitivity (recall improvement) and precision stability instead of assuming that generative models are the better models than conventional ones. Our results indicate that there is a substantial trade-off between aggressive minority detection and overall PR-AUC balance and find that SMOTE optimization outperforms CTGAN in terms of PR-AUC while CTGAN outperforms SMote in minority class recall.

II. Literature Review

A. Class Imbalance in Supervised Learning

In supervised classification tasks, one class drastically outnumbering another is a problem that is bound to occur, known as class imbalance. Even though the minority class makes up a very small portion of the data set, it represents crucial consequences in several real-world applications such as fraud detection, medical diagnosis, rare events monitoring and anomaly detection. Generally speaking, traditional machine learning algorithms are made with the implicit assumption that class distributions are balanced by maximizing overall classification accuracy. Due to this, models also often misclassify instances belonging to minorities if trained on induced datasets since they are biased towards the majority class.

The development of boundaries in decisions is affected by unequal learning. The classifier might shift the boundary towards the minority region if majority samples are more in the training process which would reduce false positives and increase false negatives. False negatives are particularly costly in high-risk areas, due to being linked to missed diagnosis, undetected fraud or system failure that is undetected. It is for these situations in which metrics for evaluation are crucial. Because a model which just predicts the majority class can still have good accuracy, accuracy in a highly unbalanced dataset is deceptive. As opposed to detection performance for minorities, and as a consistent balance between sensitivity and precision, metrics such as recall, precision, F1-score and the precision-recall area under the curve (PR-AUC) provide more insightful information.

B. Algorithm-Level Approaches

Rather than modifying the dataset, learning-time strategies modify the learning algorithm. Cost-sensitive learning is a popular technique which adjusts the misclassification penalties to increase the weight of misclassification errors of the minority classes. Class weighting mechanisms are useful to satisfy a request for the model to decrease the number of false negatives by weighting the loss function according to the class frequency. Cost-sensitive methods provide sensitivity change without changing dataset, but it does not help the minority samples to be more diverse. Limited sample diversity may reduce the generalization abilities of the classifier when the minority class has complex and multimodal possible feature distribution.

C. Generative Adversarial Networks

Two types of neural networks trained in opposition - a discriminator which tries to differentiate between generated and real data and a generator that creates synthetic samples - make up Generative Adversarial Networks. The generator is training based on adversarius training to estimate the data distribution it is used on.

The optimization problem of a minimax can be taken to represent the training objective:

$$\min_G \max_D V(D,G) = \mathbb{E}_{x\sim p_{\text{data}}}[\log D(x)] + \mathbb{E}_{z\sim p_z}[\log(1 - D(G(z)))]$$

In tasks that involve data augmentation and image synthetic tasks, GANs have displayed impressive performance. However, the presence of mixed data types, skewed distributions and multimodal features leads to challenges in the use of GANs in structured tabular data.

D. CTGAN for Tabular Data Generation

In order to overcome the drawbacks of conventional GANs when used also with tabular data, Conditional Tabular GAN (CTGAN) was created. It uses normalization methods to efficiently manage a continuous variable and conditional generation mechanisms to generate class-specific samples of them. CTGAN tries to learn the entire joint probability distribution of the data as opposed to interpolation-based methods such as SMOTE. This allows it to generate synthetic samples that take nonlinear characteristics of features into account and may be more complex and diverse. GAN-based techniques, however, introduce another one at a complexity level. As well as the need for careful hyperparameter tuning, they can be stable with regard to mode collapse or training instability, which means the generator generates samples with little diversity. In addition, compared to the conventional oversampling methods, the training of GAN algorithms can be computationally demanding.

E. Research Gap and Motivation

Despite the increasing popularity of generative models as a means to augment synthetic data, some studies have failed to employ precision-recall-focused evaluation metrics to compare the interpolation-based approaches with the GAN-based ones systematically under extreme class imbalance. Accuracy Enhancements are the subject of many of the current works, which may not accurately reflect the minority detection performance. To better understand the practical consequences of generative augmentation in very imbalanced tabular classification tasks, a comprehensive empirical comparison that emphasizes PR-AUC tradeoffs, F1-score and recall enhancement is needed.

III. Methodology

A. Problem Formulation

The problem considered in this research is binary classification under extreme class imbalance. Let us say that the dataset is represented as:

$$D = \{(x_i, y_i)\}_{i=1}^{N} \tag{1}$$

where $x_i \in \mathbb{R}^d$ denotes a feature vector of dimension d, and $y_i \in \{0,1\}$ represents the corresponding class label. In the context of fraud detection, $y = 1$ denotes fraudulent transactions (minority class), while $y = 0$ denotes legitimate transactions (majority class).

And in datasets with high class imbalance, the a priori of the minority class is vastly less than the majority class:

$$P(y=1) \ll P(y=0) \tag{2}$$

Under these circumstances, majoritarian classifiers, which are trained to minimize the total cost of classification error, have a tendency to do so in favor of the majority class. This often leads to good overall accuracy and poor performance in detecting minorities. Therefore, the major goal of this research is not to approach highest accuracy, but rather to improve the detection performance of minority classes while achieving a constant precision-recall balance. Formally, we do want to learn a classifier which maximizes minority recall and F1-score, at the same time as competitive Precision-Recall Area Under the Curve (PR-AUC).

$$f : \mathbb{R}^d \rightarrow \{0, 1\} \tag{3}$$

B. Dataset Description and Preprocessing

The experimental analysis has been performed on the dataset of Credit Card Fraud Detection, which is characterized by the extreme class imbalance issue. The dataset is made up of anonymized numeric features that were determined using the principal component analysis method, as well as a y-value representing fraudulent or legitimate transactions. The minority class will consist of less than one percent of the total observations, and so represents a good benchmark to assess imbalance-handling techniques. Since all the features are continuous and numeric, it was not necessary to do any categorical encoding.

In order to have a reliable evaluation, the dataset was divided into training and testing sets with stratified sampling. Eighty percent of the data was used for training, and the remaining twenty percent was used for the test data. Stratification maintained the original ratio of imbalance in both subsets. To avoid data leakage, all the synthetic augmentation techniques were used only for the training data. The test set was left untouched and had its original imbalance so as to simulate the real-world deployment conditions.

C. Experimental Design

The experimental framework was developed for the systematic comparison of traditional and generative methods of handling imbalances. There were three main configurations that were tested:

- A baseline model trained directly on the imbalanced dataset.
- A model trained on data balanced using SMOTE.
- A model trained on data augmented using CTGAN-generated synthetic samples.

All of the models were tested on the same test data set to try for fairness and consistency. This is a controlled design that allows for the direct comparison of performance metrics without variables that confuse the direct comparison. Additionally, class-weighted learning was explored when the impact of the way the algorithm handles imbalance was analyzed in combination with the data-level augmentation techniques.

D. Baseline Classification Model

A Random Forest classifier has been chosen as the major classification algorithm. Random Forest is a type of ensemble learning approach that builds multiple decision trees and passes them together to earn the better performance of generalization. It is especially good for structured data in the form of tabular data because it is able to fit nonlinear functions and feature interactions. The model was first trained on the unbalanced dataset to get a measure of performance. In some experimental evaluation setups, class weighting was used where more penalty is applied to minority misclassification than to majority misclassification. The class weights were automatically calculated according to inverse class frequency and thus making the minority instances more important while training the model.

E. SMOTE-Based Oversampling

In order to mitigate the imbalanced class using a classical method of the data level, the Synthetic Minority Oversampling Technique (SMOTE) was executed on the training data. SMOTE creates some synthetic minority data points by interpolating between the existing minority data instances and their nearest neighbors in the feature space.

Given a minority instance x_i and one of its nearest neighbors x_{nn}, a new synthetic sample is generated as:

$$x_{\text{new}} = x_i + \lambda(x_{nn} - x_i) \tag{4}$$

where $\lambda \in [0, 1]$ is a random interpolation factor.

This way the minority class area is widened and more samples are given to the classifier to learn from. For the next step, after creating synthetic samples, the balanced dataset was applied for re-training the Random Forest classifier, and performance was examined via the original test set.

F. CTGAN-Based Synthetic Augmentation

To get into a generative method for the handling of imbalances there is use of Conditional Tabular Generative Adversarial Networks (CTGAN). Different from SMOTE that performs interpolation for the local neighborhood, CTGAN tries to learn the joint probability distributions for the minority class. CTGAN has two neural networks, a generator and a discriminator. The generator takes random noise tensors as inputs and outputs samples, and the discriminator tries to differentiate between real samples and generated samples. The training process is adversarial with the objective of

$$\min_G \max_D V(D, G) \tag{5}$$

Through the repeated adversarial learning, the generator learns to generate the realistic samples that resemble the minority class distribution. In this study, CTGAN has been trained only on minority class examples from the training data. After training the convergence of synthetic minority samples was produced until class distribution came close to balance. These generated samples were labeled along with the original data samples which were used for training, forming an augmented dataset.

G. Evaluation Metrics

Given the gross imbalance in the data set, accuracy was not deemed a suitable measure of what is going on. Instead, performance was measured using measures that were class sensitive. Recall is a measure of the proportion of correctly detected instances of minority class:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{6}$$

Precision measures the proportion of correctly predicted positive instances among all positive predictions:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (7)$$

The F1-score provides a harmonic mean of precision and recall:

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (8)$$

In addition, the Precision-Recall Area Under the Curve (PR-AUC) was calculated to test the classifier performance for different decision thresholds. PR-AUC especially provides insight into observation of rare event detection with regards to balancing the recall with the precision.

IV. Experimental Results and Discussion

A. Quantitative Performance Evaluation

On the unaltered test dataset, the relative effectiveness of the baseline model, SMOTE-based oversampling, and CTGAN-based generative augmentation was assessed. The results are summarized in Table I.

TABLE I: Performance Comparison of Imbalance Handling Methods

Method	Recall	F1-score	PR-AUC
Baseline	0.7449	0.8391	0.8542
SMOTE	0.8265	0.8265	**0.8741**
CTGAN	**0.8571**	**0.8442**	0.8394

With a recall of 0.7449, the baseline model trained on the unbalanced dataset was able to correctly identify roughly 74.49% of fraudulent transactions. Despite its seeming reasonableness, this performance suggests that over 25% of fraud cases were overlooked, which is undesirable in high-risk applications.

B. Impact of SMOTE on Minority Representation

The application of SMOTE enhanced the recall to 0.8265 and showing that oversampling based on interpolation is useful to expand the minority class decision region. By inventing synthetic samples between the nearest neighbors, SMOTE makes the minority population more numerous, but does not actually duplicate the samples. In addition, improved recall was achieved as SMote got the maximum PR-AUC value (0.8741) which means better precision-recall stability across the classification threshold range. This implies that SMOTE not only improves the detection of minorities but also keeps them in balance regarding false positive predictions. The good PR-AUC performance shows that using interpolation-based oversampling is also very powerful for structured tabular data especially with well-behaved distributions of features and numerical transformations.

C. Effectiveness of CTGAN-Based Augmentation

CTGAN based augmentation achieved the highest recall (0.8571) and highest F1-Score (0.8442) amongst all the evaluated methods. This shows that generative augmentation improved the sensitivity of the classifier to instances from minorities. Unlike SMOTE which uses the idea of local interpolation, CTGAN tries to learn the joint probability distribution of the minority class. This enables it to create potentially more diverse synthetic samples that represent nonlinear relationships between features. However, even though it had the highest recall CTGAN achieved a lower PR-AUC (0.8394) as compared to SMote. This implies that although the model was more aggressive in the detection of fraudulent transactions, it was also more aggressive in terms of predicting false positives.

D. Comparative Interpretation

The findings show that generative augmentation is not always superior to conventional interpolation-based oversampling. Rather, the evaluation goal determines performance improvements. It is possible to make the following observations:

- Superior minority sensitivity is attained by CTGAN.
- Superior precision-recall stability is attained by SMOTE.
- Minority detection baseline performance is much worse.

These results imply that situations where minority patterns display intricate nonlinear structures might benefit more from generative models. Interpolation-based techniques, however, continue to be competitive and computationally efficient for highly transformed numerical datasets, such as PCA-based financial data.

E. Precision–Recall Curve Analysis

The Precision–Recall (PR) curve was plotted for the baseline model, SMOTE-based augmentation, and CTGAN-based augmentation in order to better examine how the assessed models behaved under various decision thresholds. Because the PR curve concentrates on minority class performance rather than overall accuracy, it is especially instructive for datasets that are highly imbalanced.

Figure 1 illustrates the tradeoff between precision and recall across different classification thresholds. Recall quantifies the percentage of real fraudulent transactions that are accurately identified, whereas precision quantifies the percentage of predicted fraudulent transactions that are in fact fraudulent.

A basic trade-off in imbalanced learning is highlighted by the PR curve analysis: increasing recall frequently leads to decreased precision. SMOTE shows more stability in preserving accuracy across thresholds, whereas CTGAN increases sensitivity to uncommon events. Thus, when weighing missed detections against false alarm rates, the choice of augmentation strategy should be in line with domain-specific operational priorities.

F. Practical Implications

The choice of imbalance handling approach should be in line with domain requirements from the standpoint of deployment:

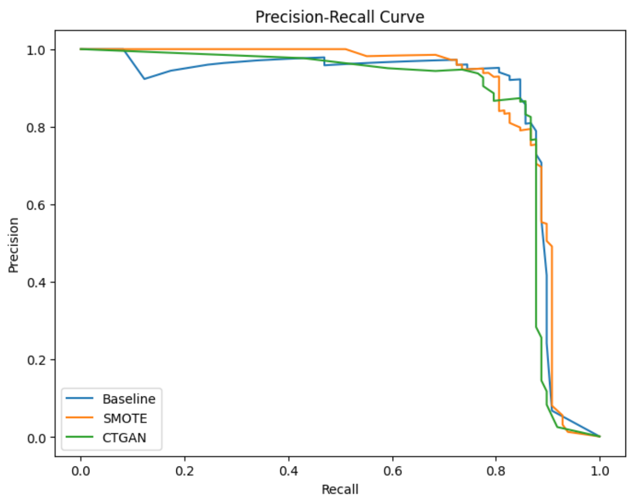

Fig. 1: Precision–Recall curves comparing Baseline, SMOTE, and CTGAN-based augmentation methods.

- CTGAN might be better if reducing missed fraud is important.
- SMOTE provides a more balanced solution if precision stability is crucial.

The findings emphasize how crucial it is to assess models using precision-recall-based metrics as opposed to depending only on accuracy gains, especially in situations with severe imbalance.

V. Conclusion and Future Work

A. Conclusion

The usefulness of generative data augmentation in severe class imbalance applications in tabular classification problems was systematically empirically studied in this work. The primary focus of this work was to find out if in the case of fraud detection, a generative model - more specifically Conditional Tabular Generative Adversarial Networks, or CT-GAN - could be found to have quantifiable advantages over standard interpolation-based oversampling techniques such as SMOTE. The experimental results indicate that without a corrective strategy, class imbalance has a significant negative effect on the detection performance for minorities. The determined limitations of conventional classifiers when facing skewed data distributions were also validated by the baseline Random Forest classifier based on the imbalanced dataset with slightly moderate recall and incapable of minority instance identification optimally.

B. Limitations and Future Directions

It is important to a certain number of limitations even though the study provides a systematic comparison between traditional and generative approaches. One financial dataset with PCA transformed numerical features as their foundation was used for the experiments. In cases with different feature structures, one would or another categorical variables, minority distributions can be linearly unseparable, the effect of generative augmentation may vary. Furthermore, although a stable configuration has been used in training the CTGAN thoroughly hyperparameter optimization was not done. It is well known that generative models can be impacted with regularization techniques, training time and architectural decision. Advanced generative architectures or different configurations may fare better and deserve to be subject to more research.

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